

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Designed and implemented **embedded C++ firmware** for enterprise IoT devices, including bare-metal and **FreeRTOS**-based application logic for microcontroller platforms with constrained memory and processing headroom; debugged and optimized code across **limited flash, RAM**, and power budgets in production environments.
- Developed **wireless communication stacks** for **Bluetooth (BLE)** and **Wi-Fi** protocols on **ESP32** microcontroller platforms using the **ESP-IDF ecosystem**; implemented device provisioning, security handshakes, and robust networking code to handle dropped connections and network recovery in IoT deployments.
- Engineered **Modbus RTU (RS-485)** and **Modbus TCP** drivers for industrial device integration and sensor interfacing; navigated register maps and bridged hardware peripherals with application logic; authored clean abstraction layers for wired protocol communication across interconnected smart home and energy devices.
- Implemented **I2C and UART communication drivers** for external hardware peripherals and real-time sensors; designed device-level integration logic and peripheral abstraction layers to ensure reliable hardware-software interfaces in production IoT ecosystems.
- Architected and deployed **cross-compilation toolchains** and **Linux-based build and test automation** using **shell scripting and Python** for firmware provisioning, configuration, and validation; managed serial/USB interfaces and wrote test harnesses for end-to-end device verification.
- Collaborated with hardware engineers, cloud backend teams, and QA to define **firmware architecture, protocol specifications, and device behaviour requirements**; conducted full firmware lifecycle management from design and implementation through production deployment, UAT coordination, and post-release monitoring via **cloud IoT infrastructure** integration.

## SKILLS

**Embedded Systems & Microcontrollers** – Embedded C/C++, bare-metal development, RTOS (FreeRTOS), ESP32 / ESP32-S/C/H series, memory management, object-oriented design patterns, constrained environments.

**Wireless Protocols & IoT** – Bluetooth (BLE), Wi-Fi (provisioning, security, TCP/IP), smart home standards, network stack implementation, connection resilience, packet debugging.

**Wired Protocols & Hardware Integration** – Modbus RTU (RS-485), Modbus TCP, I2C, UART, register maps, sensor interfacing, peripheral drivers, hardware abstraction layers.

**Firmware Development & Deployment** – ESP-IDF ecosystem, cross-compilation, toolchain management, firmware architecture, production lifecycle, UAT support, cloud IoT integration.

**Linux & Automation** – Shell scripting, Python test automation, serial/USB interface management, provisioning workflows, configuration tooling, build automation.

**Development Practices** – Git version control, CI/CD pipelines, code review, Agile/Scrum methodologies, change-control management, defect tracking, documentation.

## EDUCATION AND TRAINING

## PROJECTS

### Smart Home Energy Device Firmware (HCL Technologies)

- Architected and implemented production-grade **C++ firmware** for an **ESP32-based smart energy device** supporting **Bluetooth (BLE) and Wi-Fi connectivity** for customer provisioning and cloud synchronization.
- Engineered a **Modbus TCP driver** to integrate legacy inverter and battery-management systems; parsed and validated **register maps** for real-time energy data collection and implemented robust error handling for industrial protocol communication.
- Designed clean **abstraction layers** for **I2C sensor peripherals** (power monitoring, temperature) and **UART device interfaces**; optimized memory footprint by **40%** through static buffer pooling and efficient state-machine implementation on **FreeRTOS**.
- Developed **shell-based provisioning toolchain** and **Python test automation scripts** for firmware flashing, configuration, and end-to-end validation; integrated with **CI/CD pipelines (Jenkins)** to enable automated firmware release and deployment across device fleet.

### Embedded IoT Protocol Integration & Cloud Connectivity (HCL Technologies)

- Led firmware integration between embedded device layer and **cloud IoT infrastructure**; implemented secure **Wi-Fi and BLE communication stacks** with TLS/DTLS encryption, device authentication, and state synchronization for production smart home ecosystem.
- Collaborated with **hardware engineers** to define peripheral interface specifications (I2C, UART, SPI); with **cloud backend developers** to align firmware message contracts and device state schemas; and with **QA teams** to establish device behaviour test plans and production monitoring requirements.
- Wrote cross-platform **Linux build automation and serial debugging scripts** using **Python and shell** to streamline firmware development, reduce build-time by **35%**, and enable rapid iteration across multiple **ESP32 variants** in the product line.
- Managed full firmware lifecycle from architecture review and code implementation through UAT support, production deployment, and post-release incident response; maintained firmware reliability through systematic log analysis, telemetry collection, and iterative optimization cycles.

## ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)